

UPPSALA
UNIVERSITET

BAYESIAN HIERARCHICAL MODELS FOR INTEGRATED PROTEOMIC-PATHWAY ANALYSIS

Felix Lindström

Department of Statistics
Uppsala University

Supervisor
Måns Magnusson

2026

ABSTRACT

Contemporary techniques for estimating the effect of a treatment on biological pathways use a two-step method which fails to capture the complex interdependencies of biological systems. We propose a fully Bayesian approach which can account for these intricacies, by using hierarchical modelling and modern probabilistic computational methods, in order to draw more robust and credible inferences. Using proteomics data collected from brains of mice treated with either cocaine or saline, we specify regression-based models for observed protein intensities. The selected model uses a regularised horseshoe shrinkage prior on treatment effects, and hierarchically accounts for variation across proteins, mice, and brain regions. The hierarchical design is shown to improve predictive performance through partial pooling, and the shrinkage prior allows us to single out the pathways most affected by treatment. Compared to a two-step method, our approach finds smaller and more region-specific pathways to be most affected by treatment. Additionally, we can quantify the treatment effect on each pathway, and find that the proteins in the strongest and most credibly affected pathways have absolute intensity values shifted by up to one standard deviation due to treatment.

This thesis was written in collaboration with Erik Jansson at the Department of Pharmaceutical Biosciences, Uppsala University.

Contents

1 Introduction **1**
1.1 Related Work 2
1.2 Aim 3
1.3 Research Question 4

2 Data **4**

3 Theory **7**
3.1 Bayesian Inference 7
3.2 Hierarchical Bayesian Models 8
3.3 Hamiltonian Monte Carlo and NUTS 9

4 Method **10**
4.1 Proposed Models 10
4.2 Model Extensions 16
4.3 Convergence Diagnostics 16
4.4 Model Evaluation and Comparison 17
4.5 Baseline Two-Step Method 17

5 Results and Discussion **18**
5.1 Comparison to Two-Step Method 24

6 Conclusion **26**

Acknowledgements **27**

References **28**

Appendix A Computational Details **31**

Appendix B Stan Code **33**

Appendix C Affected Pathways **36**

Appendix D Supplementary Material **38**

1 Introduction

Life has been hypothesised to be a manifestation of matter self-organizing to accelerate the universe's degradation of free energy toward maximum entropy – the universe's most probable state (Schrödinger, 1967, pp. 71–75; England, 2013). From a different point of view, life can be thought of as a series of complex chemical reactions (Nelson et al., 2021, p. 75), which constitute a vast network of biological processes involving a myriad of different proteins (Chen et al., 2020). Proteins are essential to these processes due to their prominent role in the cells, where they function not only as the main building blocks, but also carry out nearly all the cell's functions (Alberts et al., 2022, p. 129). Proteomics is the study of the entire set of proteins in an organism, called the proteome (Anderson & Anderson, 1998), and is thus crucial for understanding how external stimuli, such as diseases or pharmaceuticals affect living beings.

In many areas of biological research, it is often of interest to quantify what impact a stimulus, for example in the form of a pharmaceutical treatment, has on a subject's biological processes, such as metabolism or the immune system. A biological process is a result of chemical reactions in a cell, involving many different proteins, where the series of reactions leading to the specific biological process is called a pathway (Schmidt et al., 2014). A simplified schematic illustrating how pathways and proteins are related is shown in Figure 1.

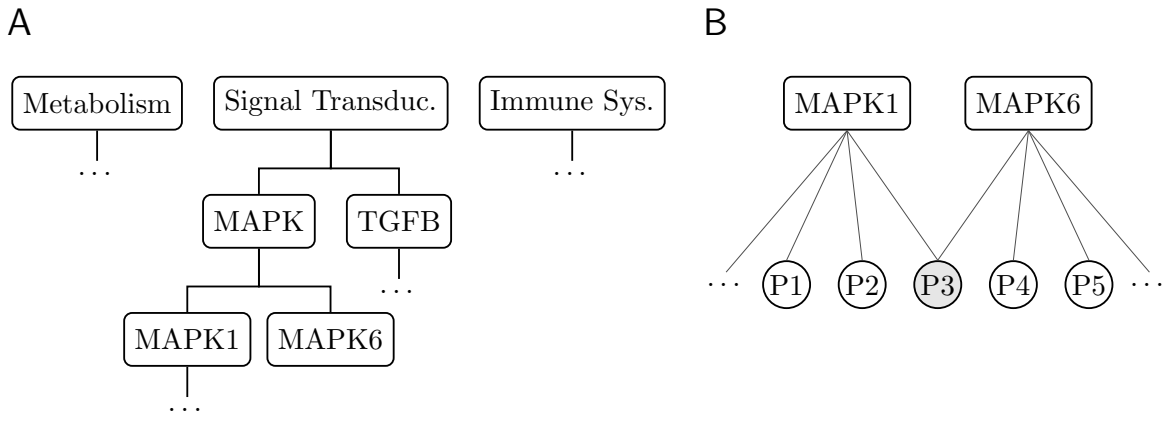


Figure 1: Example illustration of how pathways and proteins relate. Ellipses (...) indicate that there exist more connections that are not shown. **(A)** The hierarchical structure of pathways. **(B)** A demonstration of how proteins (e.g., P3) can be shared across pathways.

Pathway analysis is the statistical framework used to determine how pathways are affected by a given treatment. The analysis typically begins with obtaining protein measurements, called intensities, which can be thought of as a proxy of how much of a protein there is at the

time of measurement. Measurement is often done using a proteomics method called Liquid Chromatography-Mass Spectrometry (LC-MS) – also referred to as shotgun proteomics or bottom-up proteomics. LC breaks down proteins into smaller parts called peptides; MS then measures the peptide intensities from which the protein intensities can be inferred (Schubert et al., 2017). In pathway analysis, statistical methods are usually applied in two steps to analyse the LC-MS data. The first step, Differential Expression (DE) analysis, evaluates proteins one by one, checking for significant differences in protein intensities between treatment groups, often using t-tests (Chen et al., 2020). The second step, Enrichment Analysis, uses the results from the DE step to determine whether any pathways have been affected as a result of the proteins in that pathway having their intensity levels significantly up- or down-regulated.

Carrying out pathway analysis in two steps has considerable limitations. While biological systems are complex and made up of highly interrelated and correlated processes (Khatri et al., 2012), a two-step analysis treats the DE- and enrichment analysis-steps as disconnected, resulting in an inaccurate representation of the underlying biology. Specifically, the DE analysis does not incorporate information about which proteins are biologically involved in which pathways; the DE analysis instead considers each protein independently, and can therefore not account for more intricate protein-pathway connections.

1.1 Related Work

Proteomics datasets typically fall in the high dimensional regime, i.e., they consist of few samples on many different proteins, which leads to high variance in the protein intensity estimates. To handle this, modern inference tools often employ Empirical Bayes methods, which can stabilise variances by estimating shared hyperparameters from the data itself (Efron et al., 2001). While this is an improvement compared to frequentist t-tests, it still falls short of a fully Bayesian approach, in which the researcher places probability distributions on all unknown quantities in the model (Gelman et al., 2013, p. 101), and lets a model explore the full joint posterior distribution of the parameters. Furthermore, specifying a hierarchical structure to the fully Bayesian approach allows the model to pool information from the entire dataset to help inform individual parameter estimates (see Section 3.2), which can lead to improved parameter inference (Gelman et al., 2013, pp. 118-119).

Many tools used for proteomic analyses were initially developed not for proteins but for gene measurements. However, since they share the same challenges regarding high-dimensional data, the methods have been successfully applied to proteomics as well. Lönnstedt and Speed

(2002) were early to use empirical Bayes methods to stabilise gene variances in DE analyses with two treatment groups. Smyth (2004) generalised their results to be applicable within a linear modelling framework, enabling use of the methods for more complex research designs, and making them highly accessible.

Khatri et al. (2012) group enrichment analysis methods into two main classes. (1) Over Representation Analysis (ORA) deems a pathway enriched if the fraction of differentially expressed proteins involved in the pathway exceeds what is expected. These methods use p-value thresholds and assume independence of both proteins and pathways – an assumption which is biologically unrealistic, and can lead to biased inferences. (2) Functional Class Scoring (FCS) methods, such as `GSEA` (Subramanian et al., 2005) attempt to include more information from the DE measurements and as well as dependencies between proteins in a pathway. FCS does not require p-value thresholds, but still treats each pathway independently. Bauer et al. (2010) employed a probabilistic approach to enrichment analysis, using Markov Chain Monte Carlo (MCMC) together with a model which embedded pathways in a Bayesian network. Although their model could analyse all pathways simultaneously – a clear improvement over FCS methods – it still used the two-step paradigm, as it required DE results as input.

A general drawback of these existing methods is that while they can estimate which pathways have been affected by a treatment, and in which direction, they do not supply a treatment effect, i.e., they do not directly quantify by how much the treatment affects each pathway.

1.2 Aim

By integrating the two steps of differential expression analysis and enrichment analysis, all available information on both proteins and pathways can be combined to inform each other, leading to a more biologically realistic analysis compared to existing methods. Moreover, utilizing a fully Bayesian hierarchical approach which enables partial pooling of information offers potential for more stable inferences. To our knowledge, there is little research on how these steps can be combined; this is likely because the methods that up until now have been available, have made such an integrated approach practically infeasible due to the high computational demands. This thesis aims to implement such an integrated fully Bayesian hierarchical approach, using modern Bayesian tools such as the probabilistic programming language Stan (Stan Development Team, 2025).

1.3 Research Question

Our main objective is to investigate if pathway analysis can be successfully performed using a fully Bayesian hierarchical modelling approach, which integrates the steps of differential expression analysis and enrichment analysis. This involves determining if such an approach can detect which biological pathways are affected by treatment, and if such an approach can quantify the effect of treatment on each pathway. Additionally, we are interested in knowing how the results from using such an approach differ from when using a conventional two-step method.

2 Data

The dataset used is provided by the Department of Pharmaceutical Sciences at Uppsala University, and is the result of a randomised experiment involving six mice, where three where given treatment in the form of cocaine injection, and three were given saline (Nezhyva et al., 2024). As shown in the experimental design schematic in Figure 2, for each mouse, three repeated measurements were taken from each of $R = 4$ brain regions: Prefrontal Cortex (PFC), Nucleus Accumbens (NAc), Ventral Tegmental Area (VTA), and Hippocampus (Hip). The data thus consists of 12 samples per mouse, for a total of $N = 72$ samples. The raw samples were processed using LC-MS yielding raw intensity measurements on 2091 different proteins. Some proteins were observed in only a few samples; to avoid unstable results due to overly high variance, we follow Voghera (2023), and remove proteins observed in less than 30 samples, which leaves 1968 proteins.

As seen in Panel A in Figure 3, the raw protein intensities have a right-skewed distribution. It is standard practice in proteomics to apply a $\log_2$-transformation to the raw data, which makes the transformed distribution look approximately Normal (Panel B in Figure 3), enabling the use of a Normal likelihood when modelling. We also standardise the $\log_2$ intensities which allows us to interpret intensity deviations from 0 in terms of standard deviations. The standardised intensities are $y_{ip} = \dfrac{y^*_{ip} - \mu^*_p}{\sigma^*_p}$, where y^*_{ip} is the $\log_2$ intensity for sample i and protein p, μ^*_p is the mean $\log_2$ intensity for protein p, and σ^*_p is the $\log_2$ intensity standard deviation for protein p.

Information on which proteins are involved in which pathways is obtained from the Reactome knowledge base (Milacic et al., 2024). Reactome is an active project which includes information on all protein-pathway connections that have been discovered and annotated by

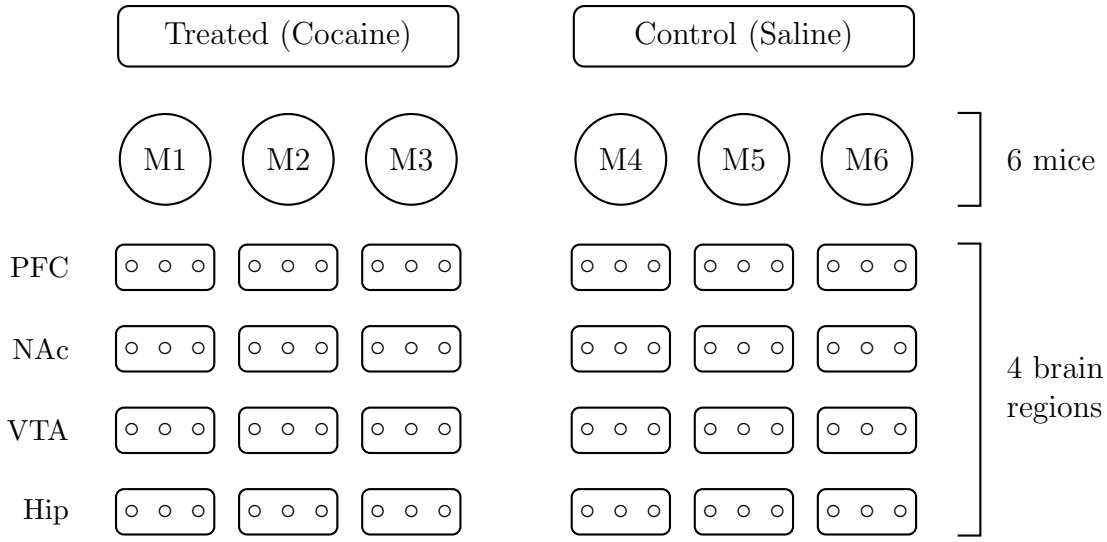


Figure 2: Overview of the experimental design. Two treatment groups each contain three animals (large circles). For each mouse, three repeated samples (circles within rectangles) were collected from each of four brain regions.

the scientific community. Reactome pathways have a hierarchical structure, meaning that small pathways lower in the hierarchy are children to larger parent pathways higher up. This implies that the proteins involved in child pathways are also involved in its parent, leading to protein overlap and correlation between pathways. This is illustrated in Figure 1 Panel A, where MAPK1 and the leaf pathway MAPK6 are both children to MAPK, so the proteins involved in them are also involved in MAPK. In general, correlation between predictors in statistical models can induce identification problems (Hansen, 2022, p. 55), which in this specific analysis means that the model has trouble distinguishing between pathways because of this overlap. To alleviate this, we include in our analysis only leaf pathways, i.e. pathways that themselves have no children. As shown by Panel B in Figure 1, protein overlap can still exist between pathways at the same level in the hierarchy. Therefore, including only leaf pathways does not fully solve the issue of overlapping pathways and identification.

Furthermore, we only include pathways which involve the proteins included in our dataset $\mathcal{D}$. Let $\mathsf{A} = \{p : p \in \mathcal{D}\}$ denote the set of unique proteins p in our dataset. Let a pathway be defined as a set of proteins $\mathsf{B}_k = \{p : p \in \text{biological process } k\}$, $(k = 1, \ldots, K)$, such that the Reactome knowledge base is a collection of pathways $\mathcal{B} = \{\mathsf{B}_1, \mathsf{B}_2, \ldots, \mathsf{B}_K\}$. The subset of pathways in Reactome which are of interest are those which share at least one protein with our dataset, and have no children. This subset is denoted as

$$\mathcal{B}_{\mathsf{A}} = \{\mathsf{B}_k \in \mathcal{B} : \mathsf{B}_k \cap \mathsf{A} \neq \emptyset \text{ and } \text{child}(\mathsf{B}_k) = \emptyset\}$$

and contains $K = 811$ pathways. The set of all proteins in the dataset that are involved in at least one of these leaf pathways is denoted as

$$\mathsf{A}_{\mathcal{B}} = \mathsf{A} \cap \Big(\bigcup_{\mathsf{B}_k \in \mathcal{B}_{\mathsf{A}}} \mathsf{B}_k \Big)$$

and contains $P = 1251$ proteins. Only a fraction of the proteins are involved in a given pathway; the average number of proteins involved in a pathway in $\mathcal{B}_{\mathsf{A}}$ is 8.

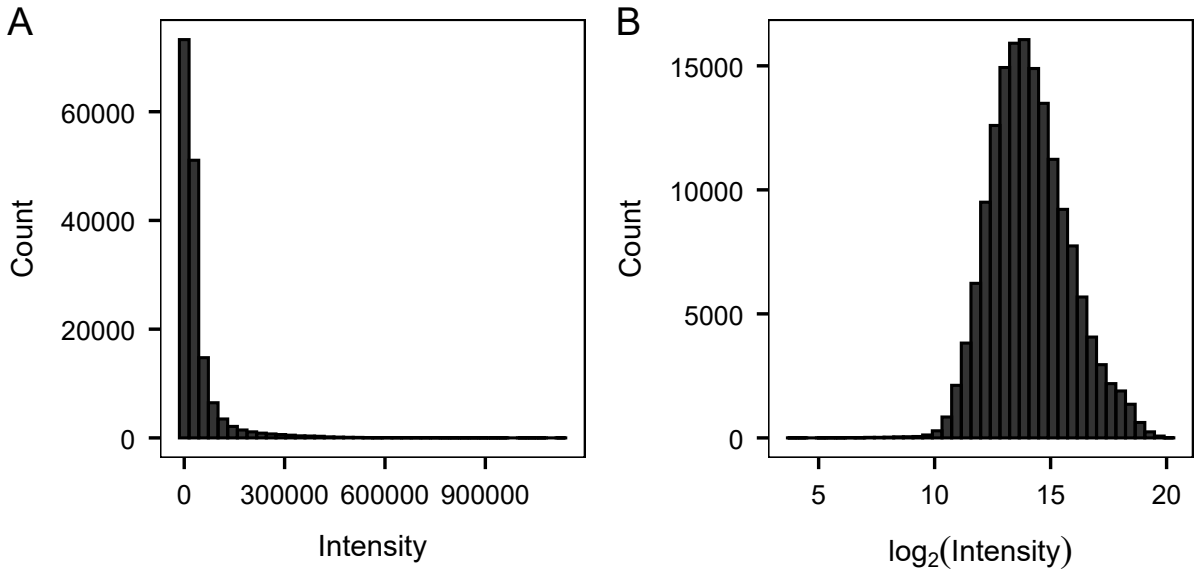


Figure 3: Distribution of the observed protein intensities. **(A):** protein intensities on the raw measurement scale. **(B):** protein intensities on the $\log_2$-scale.

For 8.2% of the retained measurements, there are no recorded values. These measurements are likely missing due to the peptide intensities being below the limit of detection (LOD) of the mass spectrometer (Karpievitch et al., 2009), which would imply that they are missing not at random. This means that the mechanism causing the missing values is non-ignorable (Gelman et al., 2013, pp. 449–451), implying that these values need to be dealt with. The observed measurements cannot be considered a random subsample of all the measurements, so excluding the missing values would cause our results to become biased (Little & Rubin, 2020, p. 13). Following Voghera (2023), we account for the missing measurements $\mathbf{y}^{\text{mis}}$ by letting Stan impute them as estimated parameters, allowing them to influence the uncertainty of the other estimated parameters of interest.

3 Theory

3.1 Bayesian Inference

With statistical inference, we aim to describe phenomena in unobserved systems, by analysing random samples of data generated by those systems. Variation in the observed data can stem from two types of uncertainty: (1) aleatoric uncertainty, i.e., irreducible randomness inherent to the phenomenon itself, or (2) epistemic uncertainty, i.e., lack of knowledge, stemming from unknown model parameters or imprecise measurements of the latent system (Betancourt, 2015; O'Hagan, 2004).

To model how observations arise from the unobserved system, we assume a probability space $(\Omega, \mathcal{F}, \mathbb{P})$, satisfying fundamental probability axioms (Kolmogorov, 1950). Here, Ω represents the space of possible outcomes, $\mathcal{F}$ is a $\sigma-$algebra of events (Tao, 2011, p. 85), and $\mathbb{P}$ is a probability measure assigning probabilities to these events (Casella & Berger, 2002, pp. 5–8). We define a random variable $Y : \Omega \to \mathcal{Y}$ which maps outcomes $\omega \in \Omega$ to observed values $y \in \mathcal{Y}$ (Klenke, 2014, pp. 1–17). This induces a probability distribution $p(\cdot)$ over the observation space $\mathcal{Y}$, often called the data-generating process (DGP), where the unknown distribution accurately characterizing the system is the true data-generating process.

Using statistical inference, we try to learn about the true distribution that governs the unobserved phenomenon, so that we can make informed decisions about that phenomenon. Since the true distribution governing the phenomenon is unknown, inference involves choosing a parametric family of candidate distributions $\mathcal{P} = \{p_{\boldsymbol{\theta}}(\cdot) : \boldsymbol{\theta} \in \boldsymbol{\Theta}\}$, and then estimating the parameters $\boldsymbol{\theta}$ from the observed data and parameter space $\boldsymbol{\Theta}$ in such a way that the chosen distribution $p_{\boldsymbol{\theta}}(\cdot)$ provides a close approximation to the true DGP (Betancourt, 2015).

Bayesian inference is a method of inference in which the observed data is considered fixed, and one's belief about the unknown true parameter values is allowed to vary. The Bayesian method updates prior knowledge of the parameters $\boldsymbol{\theta}$ with observed data $\mathbf{y} = \{y_1, \ldots, y_N\}$, using Bayes' theorem

$$p(\boldsymbol{\theta} \mid \mathbf{y}) = \frac{p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})}{\int p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta})\, d\boldsymbol{\theta}} \;\propto\; p(\mathbf{y} \mid \boldsymbol{\theta})\, p(\boldsymbol{\theta}),$$

where $p(\boldsymbol{\theta})$ is the prior distribution, $p(\mathbf{y} \mid \boldsymbol{\theta})$ is the likelihood of the data, and $p(\boldsymbol{\theta} \mid \mathbf{y})$ is the posterior distribution of the parameters given the observed data. This contrasts the frequentist method, which assumes random data, and uses limit frequencies of events to quantify uncertainty about unknown but fixed parameters.

3.2 Hierarchical Bayesian Models

Hierarchical Bayesian models are well-suited to settings where observed units are neither independent nor identical, but can instead be regarded as exchangeable draws from a common population (Bernardo, 1996). By organising parameters in a hierarchy, the model allows units to share statistical strength while still permitting between-unit variation (Gelman et al., 2013, p. 101).

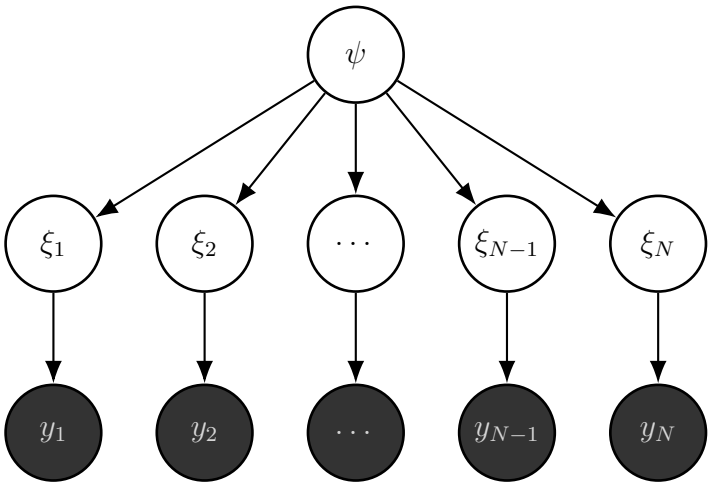


Figure 4: Example of a one-level hierarchical model with lower-level parameters ξ_i and a shared hyperparameter ψ.

A simple one-level hierarchy with parameters $\boldsymbol{\theta} = (\boldsymbol{\xi}\, \psi)$, where $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_N,)$, and observed data $\mathbf{y} = (y_1, \ldots, y_N)$ is illustrated in Figure 4. Lower-level parameters ξ_i are assumed to arise from a common prior distribution $p(\xi_i \mid \psi)$, conditional on the higher-level parameter ψ, commonly referred to as a hyperparameter, which is itself assigned a hyperprior distribution $p(\psi)$. Conditional on the lower-level parameters, observed data are generated from the likelihood $p(y_i \mid \xi_i)$. Assuming independent observations, the joint posterior factors as

$$p(\boldsymbol{\xi}, \psi \mid \mathbf{y}) \propto \prod_{i=1}^{N} p(y_i \mid \xi_i)\, p(\xi_i \mid \psi)\, p(\psi).$$

In Figure 4, it is clear that the hyperparameter ψ influences the generation of each ξ_i, leading to a dependence between units i. This dependence means that even though each data point y_i only directly depends on its corresponding parameter ξ_i, all observed values contribute to the posterior of ψ, which in turn regularises all ξ_i toward a common value. This hierarchical mechanism gives rise to what is called partial pooling (Betancourt & Girolami, 2013), where the observed data points share information to inform the estimated parameters.

Hierarchical models fall between two non-hierarchical extremes: no pooling, where parameters are estimated independently, and complete pooling, where all data points are assumed to share a single common parameter.

3.3 Hamiltonian Monte Carlo and NUTS

In statistical inference, interest often lies in computing expectations of functions of parameters

$$\mathbb{E}[f(\boldsymbol{\theta})] = \int f(\boldsymbol{\theta})\, p(\boldsymbol{\theta} \mid \mathbf{y})\, d\boldsymbol{\theta}. \tag{1}$$

However, computing the integral in (1) becomes infeasible when the parameter space is high-dimensional and the posterior distribution is complex (Betancourt, 2018; Hoffman & Gelman, 2014). A solution is to use sampling algorithms, such as Markov Chain Monte Carlo (MCMC) to collect samples of the parameters $\boldsymbol{\theta}$ from the posterior distribution which can be used to approximate the expectation.

While traditional MCMC algorithms have proved ineffective when posterior distributions become complex (Neal, 2011), the Hamiltonian Monte Carlo (HMC) algorithm was developed to exploit the geometry of the posterior distribution's high-mass areas, called the typical set (Betancourt, 2018), to efficiently explore the posterior distribution (Roberts & Rosenthal, 2004). HMC adds auxiliary momentum variables $\boldsymbol{\phi}$ to the posterior distribution parameters $\boldsymbol{\theta}$, creating the joint density

$$\pi(\boldsymbol{\theta}, \boldsymbol{\phi}) = \pi(\boldsymbol{\phi} \mid \boldsymbol{\theta})\, \pi(\boldsymbol{\theta}).$$

Here, $\pi(\cdot)$ denotes a distribution, and $\pi(\boldsymbol{\theta}) = p(\boldsymbol{\theta} \mid \mathbf{y})$ is the posterior distribution. The joint density can be written in terms of the Hamiltonian

$$\begin{aligned} H(\boldsymbol{\theta}, \boldsymbol{\phi}) &= -\log \pi(\boldsymbol{\theta}, \boldsymbol{\phi}) \\ &= -\log \pi(\boldsymbol{\phi} \mid \boldsymbol{\theta}) - \log \pi(\boldsymbol{\theta}) \\ &=: K(\boldsymbol{\phi} \mid \boldsymbol{\theta}) + U(\boldsymbol{\theta}) \\ &= [\boldsymbol{\phi}^\top M^{-1} \boldsymbol{\phi}/2] + [-\log \pi(\boldsymbol{\theta})], \end{aligned}$$

where $K(\boldsymbol{\phi} \mid \boldsymbol{\theta})$ is called the kinetic energy, $U(\boldsymbol{\theta})$ is called the potential energy, and $H(\boldsymbol{\theta}, \boldsymbol{\phi})$ is the total energy of the system (Neal, 2011). The HMC algorithm can then be guided through the typical set of the posterior distribution (Betancourt & Girolami, 2013), leading to efficient exploration of the posterior distribution, using gradient information from Hamilton's

equations,

$$\frac{d\boldsymbol{\theta}}{dt} = +\frac{\partial H(\boldsymbol{\theta}, \boldsymbol{\phi})}{\partial \boldsymbol{\phi}} = \frac{\partial [U(\boldsymbol{\theta}) + K(\boldsymbol{\phi} \mid \boldsymbol{\theta})]}{\partial \boldsymbol{\phi}} = \frac{\partial K(\boldsymbol{\phi} \mid \boldsymbol{\theta})}{\partial \boldsymbol{\phi}} = \frac{\partial \left[\boldsymbol{\phi}^\top M^{-1} \boldsymbol{\phi}/2\right]}{\partial \boldsymbol{\phi}} = M^{-1}\boldsymbol{\phi},$$
$$\frac{d\boldsymbol{\phi}}{dt} = -\frac{\partial H(\boldsymbol{\theta}, \boldsymbol{\phi})}{\partial \boldsymbol{\theta}} = -\frac{\partial [U(\boldsymbol{\theta}) + K(\boldsymbol{\phi} \mid \boldsymbol{\theta})]}{\partial \boldsymbol{\theta}} = -\frac{\partial [U(\boldsymbol{\theta})]}{\partial \boldsymbol{\theta}} = \frac{\partial \log \pi(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}$$

where M is a symmetric and positive definite mass matrix. Evolving this Hamiltonian system over discretized time steps t creates a trajectory in phase space (Strogatz, 2024, p. 8). At the end of an iteration – when all steps have been taken – a parameter sample $\boldsymbol{\theta}$ is drawn from this trajectory. In practice, an extension of HMC called the No-U-Turn Sampler (NUTS) (Hoffman & Gelman, 2014) is used, which adaptively sets the number of steps to take and automatically tunes the step-size used when evolving the Hamiltonian system.

4 Method

The following models implement a fully Bayesian framework using the probabilistic programming language Stan and the `rstan` package (Stan Development Team, 2025) in `R`. We aim to understand how the treatment affects the biological pathways, by modelling how those effects manifest in the observed protein intensity values. We start by specifying a simple model, and then increase the complexity of subsequent models. For simulation details, such as number of chains, iterations, and NUTS-parameters used, see Table 3 in Appendix A.

4.1 Proposed Models

Model 1

$$\begin{aligned}
y_{ip}^{\text{obs}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
y_{ip}^{\text{mis}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
\mu_{ip} &= \alpha_{apr} + w_i \sum_{k=1}^{K} x_{kp}\, \delta_{kr} \\
\alpha_{apr} &\sim \text{Normal}\,(0, 1) \\
\delta_{kr} &\sim \text{Normal}\,(0, 1) \\
\sigma_p &\sim \text{Exponential}\,(1) \\
y_{ip}^{\text{mis}} &\sim \text{Normal}\left(\min(\mathbf{y}_p^{\text{obs}}),\, 1\right).
\end{aligned} \tag{2}$$

In Model 1, y_{ip}^{obs} is the observed intensity for protein $p \in \{1, \ldots, P\}$ and sample $i \in \{1, \ldots, N\}$. y_{ip}^{mis} and μ_{ip} are the corresponding missing value intensity and expected intensity, respectively. σ_p^2 is the intensity residual variance for protein p, capturing variation unexplained by μ_{ip}. α_{apr} is the baseline intensity for animal $a \in \{1, \ldots, 6\}$, protein p, and brain region $r \in \{1, \ldots, 4\}$, where a denotes the animal that sample i was taken from, and r denotes the brain region that sample i was taken from. The indexing over animals intends to capture between-mouse variation in baseline intensity, motivated by the fact that the data was collected using repeated measurements on the same six mice. δ_{kr} is the treatment effect for pathway $k \in \{1, \ldots, K\}$ and brain region r, and $\boldsymbol{\delta}$ is the vector of all such treatment effects. w_i is an indicator taking the value 1 if sample i is treated, and 0 otherwise. $\mathbf{X} \in \mathbb{R}^{K \times P}$ is a pathway mapping matrix, mapping proteins to pathways via the known biological relationships in Reactome, where x_{kp} denotes its (k, p)-th element, such that

$$x_{kp} = \begin{cases} 1 & \text{if protein } p \text{ belongs to pathway } k, \\ 0 & \text{otherwise.} \end{cases}$$

We recognise that since $\mathbf{X}$ is a binary matrix, treatment effects δ_{kr} for larger pathways involving a larger number of proteins are represented more often in the full likelihood compared to smaller pathways, causing treatment effects for larger pathways to be informed by disproportionately more data. We attempted to balance this by scaling $\mathbf{X}$ by $1/\sqrt{|k|}$, where $|k|$ is the pathway size. However, using this scaling to fit the model severely degraded predictive performance, which led us to abandon this approach.

The model can be viewed as a linear regression, with α_{apr} as the intercept, w_i as the explanatory variable, and $\sum_{k=1}^{K} x_{kp}\, \delta_{kr}$ as the regression coefficient. The model uses a Normal likelihood, which seems to be a reasonable assumption for the log-transformed data (Panel B, Figure 3). For the control samples, the expected intensity is simply the intercept. For treated samples, the term $w_i \sum_{k=1}^{K} x_{kp}\, \delta_{kr}$ is added, which encodes the treatment effect on protein p as the sum of the effects on all pathways containing that protein. Conversely, δ_{kr} represents the partial treatment effect on protein p, attributable specifically to pathway k in region r.

The model considers missing values y_{ip}^{mis} by sampling them as parameters, and evaluating them in a separate but identical likelihood. By doing this we let the missing values inform the joint posterior of all the other parameters of interest. Following (Voghera, 2023), we use a Normal prior centred around the smallest observed value for each protein, $\min(\mathbf{y}_p^{\text{obs}})$, reflecting the assumption that missing values arise from low-intensity proteins below the

LOD, where $\mathbf{y}_p^{\text{obs}}$ is the vector of all observed intensities for protein p. The minimum observed value per protein has a median of -2.47 on the standardised scale ($\min = -7.97, \mathrm{q}_{25} = -3.02, \mathrm{q}_{75} = -2.06, \max = -1.04$), confirming that these values lie well below the mean intensity.

The priors on the other parameters are intended to be weakly informative relative to the scale of the standardised data. The standard Normal priors on α_{apr} and δ_{kr} allow both positive and negative effects, and provide mild regularisation to avoid extreme estimates. Because the data is standardised before modelling, a prior mean of 0 reflects the global average of the protein intensities. The Exponential(1) prior on the residual standard deviation σ_p guarantees positive estimates with a mean of 1 while still allowing moderate variability.

Model 2

$$
\begin{aligned}
y_{ip}^{\text{obs}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
y_{ip}^{\text{mis}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
\mu_{ip} &= \alpha_{apr} + w_i \sum_{k=1}^{K} x_{kp}\, \delta_{kr} \\
\alpha_{apr} &\sim \text{Normal}\left(0, \tau_\alpha^2\right) \\
\delta_{kr} &\sim \text{Normal}\left(0, 1\right) \\
\sigma_p &\sim \text{LogNormal}\left(0, \tau_\sigma^2\right) \\
\tau_\alpha &\sim \text{Student-}\, t^+\left(3, 0, 1\right) \\
\tau_\sigma &\sim \text{Student-}\, t^+\left(3, 0, 1\right) \\
y_{ip}^{\text{mis}} &\sim \text{Normal}\left(\min(\mathbf{y}_p^{\text{obs}}),\, 1\right).
\end{aligned}
\tag{3}
$$

Model 2 introduces a hierarchical scale structure on the Normal prior on α_{apr}, and on the LogNormal prior on σ_p, by defining the hierarchical scale parameters τ_α and τ_σ. The log scale is used to ensure positive estimates for the protein variances. While the priors in Model 1 are static, the new hierarchical structure in Model 2 allows estimates of baseline protein intensities α_{apr} and residual variances σ_p^2 for individual proteins p to be partially informed by the rest of the observed proteins, leading to more stable inferences.

The newly defined prior scale parameters are in turn given Student's t hyperpriors which are meant to be weakly informative while allowing larger variation in baseline intensities and residual variance.

Model 3

Model 3, in (4), places a horseshoe prior (Carvalho et al., 2009) on the treatment effects δ_{kr}. We use a slightly modified horseshoe prior, where the Cauchy$(0, 1)$ priors of Carvalho et al. (2009) are replaced with half-Student's t priors, and the global shrinkage scale is calibrated via τ_0 following Piironen and Vehtari (2017).

$$
\begin{aligned}
y_{ip}^{\text{obs}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
y_{ip}^{\text{mis}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
\mu_{ip} &= \alpha_{apr} + w_i \sum_{k=1}^{K} x_{kp}\, \delta_{kr} \\
\alpha_{apr} &\sim \text{Normal}\left(0, \tau_\alpha^2\right) \\
\delta_{kr} \mid \lambda_{kr}, \tau_r &\sim \text{Normal}\left(0, \tau_r^2 \lambda_{kr}^2\right) \\
\lambda_{kr} \sim \text{Student-}\, t^+\,(3, 0, 1)\,, &\quad \tau_r \sim \text{Student-}\, t^+\,(3, 0, \tau_0) \\
\sigma_p &\sim \text{LogNormal}\left(0, \tau_\sigma^2\right) \\
\tau_\alpha &\sim \text{Student-}\, t^+\,(3, 0, 1) \\
\tau_\sigma &\sim \text{Student-}\, t^+\,(3, 0, 1) \\
y_{ip}^{\text{mis}} &\sim \text{Normal}\left(\min(\mathbf{y}_p^{\text{obs}}),\, 1\right).
\end{aligned}
\tag{4}
$$

Under the horseshoe prior, δ_{kr} is assigned a conditional Normal distribution with hierarchical scale parameters λ_{kr} and τ_r. When $\text{Var}(\delta_{kr} \mid \lambda_{kr}, \tau_r) = \tau_r^2 \lambda_{kr}^2$ is small, the Normal prior on δ_{kr} is narrowly concentrated around 0, effectively "shrinking" that parameter towards 0. τ_r is a global shrinkage parameter, controlling the general shrinkage for all δ_{kr} in brain region r. λ_{kr} is a local shrinkage parameter for each pathway k and brain region r; if this parameter is large, δ_{kr} can escape the global shrinkage effect of τ_r. The functionality of allowing shrinkage to be either applied or not when necessary, is enabled by the heavy-tailed half Student's t priors on the shrinkage parameters. A prior guess of the number of non-zero δ_{kr}-parameters per brain region, denoted p_0, is used to calibrate the global shrinkage variance: $\tau_0 = \dfrac{p_0}{K - p_0} \dfrac{\sigma}{\sqrt{N}}$. The outcome variable scale parameter σ is set to 1 because we use standardised data, and we consider three different sparsity levels by setting $p_0 \in \{3, 10, 20\}$.

A shrinkage prior such as the horseshoe is useful in situations where the underlying signal is sparse, meaning that most of the parameters are negligibly small. We believe this sparsity

assumption to hold for our treatment effect parameters δ_{kr}, because it is plausible that only a few pathways within each brain region are significantly affected by the treatment.

Model 4

Model 4, in (5), places a regularised horseshoe (RHS) prior (Piironen & Vehtari, 2017) on the treatment effects δ_{kr}. While the original horseshoe prior is, in certain cases, susceptible to identification problems, causing parameter estimates to diverge to infinity, the RHS attempts to solve this problem by shrinking even the largest-signal parameters by at least a small amount. Our proteomics data is such a case where many of the Reactome pathways overlap with respect to the proteins involved in them, potentially causing identifiability problems.

$$
\begin{aligned}
y_{ip}^{\text{obs}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
y_{ip}^{\text{mis}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
\mu_{ip} &= \alpha_{apr} + w_i \sum_{k=1}^{K} x_{kp}\, \delta_{kr} \\
\alpha_{apr} &\sim \text{Normal}\left(0, \tau_\alpha^2\right) \\
\delta_{kr} \mid \lambda_{kr}, \tau_r, c &\sim \text{Normal}\left(0, \tau_r^2 \tilde{\lambda}_{kr}^2\right) \\
\tilde{\lambda}_{kr}^2 = \frac{c^2 \lambda_{kr}^2}{c^2 + \tau_r^2 \lambda_{kr}^2}, &\quad \lambda_{kr} \sim \text{Student-}\, t^+\,(3, 0, 1) \\
\tau_r \sim \text{Student-}\, t^+\,(3, 0, \tau_0)\,, &\quad c^2 \sim \text{Inv Gamma}\,(5, 5) \\
\sigma_p &\sim \text{LogNormal}\left(0, \tau_\sigma^2\right) \\
\tau_\alpha &\sim \text{Student-}\, t^+\,(3, 0, 1) \\
\tau_\sigma &\sim \text{Student-}\, t^+\,(3, 0, 1) \\
y_{ip}^{\text{mis}} &\sim \text{Normal}\left(\min(\mathbf{y}_p^{\text{obs}}),\, 1\right).
\end{aligned}
\tag{5}
$$

In the RHS, the term $\tilde{\lambda}_{kr}$ works in the following way to both facilitate shrinkage on small parameters while limiting the variance of large parameters. When a parameter δ_{kr} is near 0, i.e., $\tau_r^2 \lambda_{kr}^2 \ll c^2$, then $\tilde{\lambda}_{kr}^2 \to \lambda_{kr}^2$, making the RHS approach the original horseshoe. When a parameter is far from 0, i.e., $\tau_r^2 \lambda_{kr}^2 \gg c^2$, then $\tilde{\lambda}_{kr}^2 \to c^2/\tau_r^2$ making the RHS approach a Normal$(0, c^2)$. Thus, while the original horseshoe allows arbitrarily large parameter values, the c^2 parameter in the RHS limits the estimated parameters' variance. Following Piironen

and Vehtari (2017), an Inverse Gamma(α, β) with $\alpha = \nu/2$ and $\beta = \nu s^2/2$ is placed on c^2 as a weakly informative prior, where ν are the degrees of freedom. This translates to a Student's $t(\nu, \mu, s^2)$, such that with $\alpha = 5$ and $\beta = 5$, the parameters far from 0 approximately follow a Student's $t(10, 0, 1)$ distribution. Stan code implementing this model can be found in Appendix B.

Model 5

Model 5, in (6) extends Model 4 by introducing a hierarchical structure on the global shrinkage parameters τ_r, while keeping the rest of the model unchanged. By introducing the scale hyperparameter κ, the global shrinkage parameters for the regions r are allowed to share information in the estimation process through partial pooling. This is motivated by the notion that pathway effects in different brain regions may not be independent, but connected through some latent process.

$$
\begin{aligned}
y_{ip}^{\text{obs}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
y_{ip}^{\text{mis}} \mid \alpha_{apr}, \boldsymbol{\delta}, \sigma_p &\sim \text{Normal}\left(\mu_{ip}\,,\, \sigma_p^2\right) \\
\mu_{ip} &= \alpha_{apr} + w_i \sum_{k=1}^{K} x_{kp}\, \delta_{kr} \\
\alpha_{apr} &\sim \text{Normal}\left(0, \tau_\alpha^2\right) \\
\delta_{kr} \mid \lambda_{kr}, \tau_r, c &\sim \text{Normal}\left(0, \tau_r^2 \tilde{\lambda}_{kr}^2\right) \\
\tilde{\lambda}_{kr}^2 = \frac{c^2 \lambda_{kr}^2}{c^2 + \tau_r^2 \lambda_{kr}^2}, &\quad \lambda_{kr} \sim \text{Student-}\, t^+\left(3, 0, 1\right) \\
\tau_r \sim \text{Student-}\, t^+\left(3, 0, \kappa\right), &\quad \kappa \sim \text{Student-}\, t^+\left(3, 0, \tau_0\right), \quad c^2 \sim \text{Inv Gamma}\left(5, 5\right) \\
\sigma_p &\sim \text{LogNormal}\left(0, \tau_\sigma^2\right) \\
\tau_\alpha &\sim \text{Student-}\, t^+\left(3, 0, 1\right) \\
\tau_\sigma &\sim \text{Student-}\, t^+\left(3, 0, 1\right) \\
y_{ip}^{\text{mis}} &\sim \text{Normal}\left(\min(\mathbf{y}_p^{\text{obs}}),\, 1\right).
\end{aligned}
\tag{6}
$$

4.2 Model Extensions

As extensions to the five proposed models, we investigate how two different changes to the baseline protein intensity (intercept) term affects model performance. The first extension adds a hierarchical location parameter μ_α which encourages partial pooling towards a shared location rather than towards zero.

Extension 1

$$\begin{aligned} \alpha_{apr} &\sim \text{Normal}\left(\mu_\alpha, \tau_\alpha^2\right) \\ \mu_\alpha &\sim \text{Normal}\left(0, 1\right) \\ \tau_\alpha &\sim \text{Student-}\, t^+\left(3, 0, 1\right). \end{aligned} \tag{7}$$

The second extension implements a linear specification of the baseline intensities with separate hierarchical scale parameters across animals, proteins, and brain regions, which greatly reduces the number of model parameters.

Extension 2

$$\begin{gathered} \alpha_{apr} = \alpha_a + \alpha_p + \alpha_r \\ \alpha_a \sim \text{Normal}\left(0, \tau_a^2\right),\ \alpha_p \sim \text{Normal}\left(0, \tau_p^2\right),\ \alpha_r \sim \text{Normal}\left(0, \tau_r^2\right) \\ \tau_a,\ \tau_p,\ \tau_r \sim \text{Student-}\, t^+\left(3, 0, 1\right). \end{gathered} \tag{8}$$

Here, α_a, α_p and α_r represent the effects for animal a, protein p, and brain region r respectively, controlled by their corresponding hierarchical scale parameters τ_a, τ_p, and τ_r.

4.3 Convergence Diagnostics

To be able to make valid inferences using a sampling algorithm like HMC, it must be ensured that the simulated parameter samples are representative of the posterior distribution. The multiple Markov chains are said to have converged, if, after a certain number of iterations, they have mixed together sufficiently to explore the same area of the typical set (Gelman et al., 2013, pp. 284–285). Convergence for each parameter is typically evaluated using $\hat{R} \in [1, \ldots, \infty]$, and Effective Sample Size (ESS) which measures how many independent samples the sampling algorithm has produced. To obtain stable inferences for quantities of interest Vehtari et al. (2021) recommend that all $\hat{R} < 1.01$, and ESS > 100 per chain.

4.4 Model Evaluation and Comparison

To compare the performance of different models, the fitted models' predictive performances can be compared. The Pareto-smoothed importance-sampling leave-one-out (PSIS-LOO) estimate of the expected log pointwise predictive density (ELPD) has been shown by Vehtari et al. (2017) to be a robust and computationally efficient estimate of predictive performance. The quantity is computed as

$$\widehat{\text{elpd}}_{\text{psis-loo}} = \sum_{i=1}^{n} \log \left[\frac{\sum_{s=1}^{S} w_i^s \, p(y_i \mid \theta^s)}{\sum_{s=1}^{S} w_i^s} \right],$$

where $p(y_i \mid \theta^s)$ is the likelihood for observed value i given parameters θ from posterior draw $s \in \{1, \ldots, S\}$. w_i^s are Pareto-smoothed importance weights, which are used to efficiently approximate leave-one-out cross-validation. The reliability of the ELPD estimates are assessed using the Pareto shape parameter $\hat{k}$. If $\hat{k} > 0.7$ for an observed value, the variance of that ELPD estimate may be large and the estimate thus unreliable (Vehtari et al., 2017, p. 1416). When comparing fitted models, a higher ELPD estimate is indicative of better predictive performance.

4.5 Baseline Two-Step Method

To provide a baseline that we can compare our proposed Bayesian approach to, we conduct pathway analysis using a conventional two-step method. Differential expression analysis is performed using `limma` (Ritchie et al., 2015), which fits a separate linear model per protein to estimate the treatment log-fold change, and applies an empirical Bayes procedure to stabilise variance estimates across proteins, yielding a moderated t-statistic for each protein. Proteins are then ranked by their t-statistic, and this ranked list is used as input to `fgsea` (Korotkevich et al., 2016), an FCS enrichment analysis method that evaluates whether proteins within each Reactome pathway are disproportionately concentrated at the extremes of the ranked list. Pathway significance is then summarised using the False Discovery Rate (FDR), which is the proportion of false positives among the positives, i.e., the number of times the pathway was falsely found to be enriched out of the total times the method found it to be enriched.

We use the same dataset and leaf pathways as for the fully Bayesian models. `limma` includes a separate intercept and treatment effect for each brain region, and to account for repeated measurements from animals, average within-animal correlation across proteins is estimated. Before fitting the model, missing values are imputed with the minimum value per protein, as opposed to being estimated as parameters in the fully Bayesian approach.

5 Results and Discussion

Sampling converged adequately for all models, with the majority of $\hat{R}$ values close to 1.000, and the majority of parameters having ESS > 100 per chain. This means that we can proceed to make inferences using these models, knowing that they will be valid. See Table 5 in Appendix A for a summary of $\hat{R}$ and ESS values.

Table 1: Difference in predictive performance for each model, relative to the model with the highest estimated ELPD score. S.E. = Standard Error.

		$\Delta\,\widehat{\text{elpd}}_{\text{psis-loo}}$	S.E.
Model 4	($p_0 = 10$)	0.0	0.0
Model 4	($p_0 = 3$)	−3.1	15.1
Model 3	($p_0 = 10$)	−3.9	15.1
Model 3	($p_0 = 20$)	−10.0	15.3
Model 5	($p_0 = 20$)	−10.6	15.7
Model 3	($p_0 = 3$)	−13.9	15.3
Model 5	($p_0 = 3$)	−14.6	15.3
Model 5	($p_0 = 10$)	−18.3	15.0
Model 4	($p_0 = 20$)	−20.0	14.8
Model 2		−93.5	23.1
Model 1		−230.4	30.6

Results concerning predictive performance of the models are shown in Table 1. We note that for each model, $\hat{k} > 0.7$ for roughly 3-5% of the observed values, implying that ELPD estimates may be unreliable. With this in mind, results in Table 1 shows that ELPD for the models that use a shrinkage prior on the treatment effects outperform the other two models. This aligns with the expectation of sparse treatment effects, where the shrinkage prior successfully isolates strong signals while suppressing noise. Differences in ELPD among the shrinkage prior models are however well within two standard errors, indicating that their performance is similar. Given the overlapping structure of pathways and the risk of identifiability problems that it may induce, we argue that a model with the regularised horseshoe is a more robust choice compared to a model using the original horseshoe prior. Moreover, since the addition of hierarchical variance on global shrinkage in Model 5 provided no meaningful improvement in predictive performance, we select the simpler Model 4 as our preferred model. Furthermore, varying the sparsity levels $p_0 \in \{3, 10, 20\}$ did not have a strong impact on predictive performance. We therefore choose the $p_0 = 3$ setting because

it reflects the most conservative assumption that relatively few of the 811 pathways are genuinely affected by treatment.

Table 2: (Left): Number of pathways with large treatment effect. **(Right):** Number of pathways with at least 95% posterior probability of being either positive or negative.

	# pathways with $\lvert\text{posterior mean}(\delta_{kr})\rvert > 0.5$				# pathways with $\Pr(\delta_{kr} > 0)$ or $\Pr(\delta_{kr} < 0) \geq 0.95$			
	PFC	NAc	VTA	Hip	PFC	NAc	VTA	Hip
Model 1	21	55	84	58	7	14	27	23
Model 2	27	78	101	76	15	30	50	37
Model 3	0	1	1	0	2	2	4	2
Model 4	0	1	1	0	2	2	4	2
Model 5	0	1	1	0	3	2	4	2

The model extensions presented in Section 4.2 were implemented and fitted as revised specifications of Model 4 with $p_0 = 3$, and compared to Model 4 as specified in (5). The results in Table 4 in Appendix A show that neither of the extensions resulted in an improvement in predictive performance compared to the unextended Model 4. The ELPD for Extension 1 of -98.4 is well outside two standard deviations (15.6), indicating that the introduced hierarchical location parameter μ_α conflicts with the α_{apr} specification. Extension 2 receives an ELPD of -42455.2 (SE $= 291.3$), which shows that what was gained in terms of simpler model specification was severely lost in predictive performance. The $\hat{k}$ parameters for extension 1 showed the same pattern as the unextended models; for extension 2 all $\hat{k}$ were below 0.7.

The effect of the horseshoe and regularised horseshoe priors can be seen in Table 2. The left panel shows for how many pathways, treatment shifted the posterior mean of the expected protein intensity by at least 0.5 standard deviations for the proteins belonging to that pathway, while holding all other pathway effects constant. The right panel shows for how many pathways the posterior probability of the treatment effect being either strictly positive or strictly negative is at least 95%, indicating a credible direction of effect. Models 3, 4, and 5, which employ shrinkage priors, find far fewer strongly affected pathways, and far fewer credibly affected pathways, compared to models 1 and 2. Taken together, the results illustrate that the shrinkage priors are functioning as intended; i.e., by shrinking ambiguous effects to zero, the number of credibly affected pathways is condensed, which potentially reduces the number of false positives.

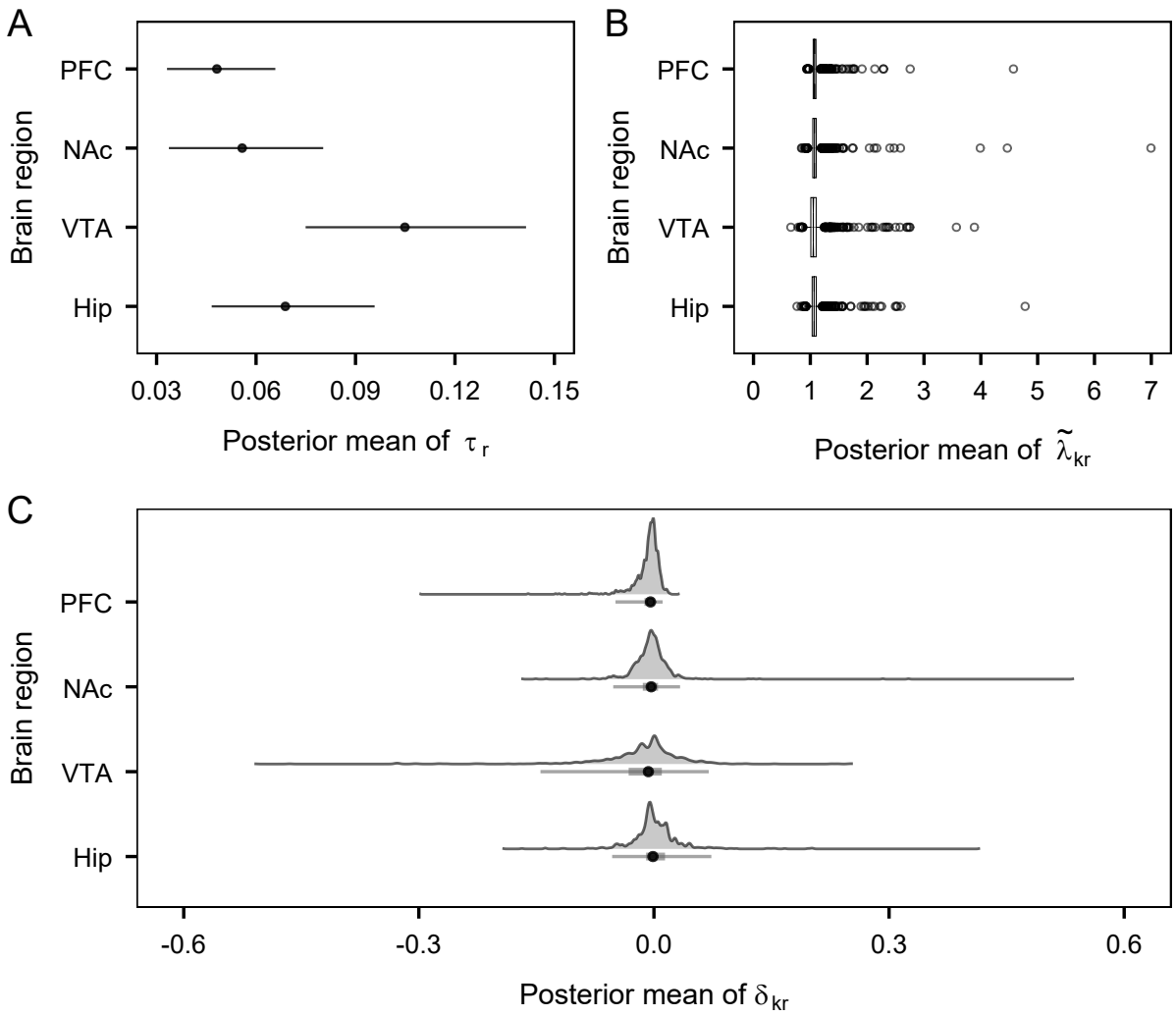


Figure 5: Regularised horseshoe parameter results for Model 4. **(A):** Posterior mean and 95% credible intervals for the global shrinkage parameter τ_r. **(B):** Box plots of the posterior mean of the local shrinkage parameter $\tilde{\lambda}_{kr}$. **(C):** Posterior mean distributions of the pathway treatment effects δ_{kr}. The black point represents the median; grey lines are 50% and 90% credible intervals.

Focusing on our preferred model (Model 4), the shrinkage pattern of the regularised horseshoe can be further observed in Figure 5. While there is some variation in the distribution of the global shrinkage parameter τ_r (Panel A), with the shrinkage in VTA being relatively weaker, the global shrinkage toward zero is generally quite strong. The distribution of the local shrinkage parameter $\tilde{\lambda}_k$ for each pathway k, in Panel B, shows the median local shrinkage close to one for each brain region. However, as shown by the points to the right of the boxes, a substantial number of pathways have local shrinkage parameters greater than $1.5 \times$ IQR (interquartile range). Despite this, the global shrinkage is strong enough to keep most treatment effects close to zero, as is evident by the posterior mean distributions of δ_{kr} in

Panel C of Figure 5.

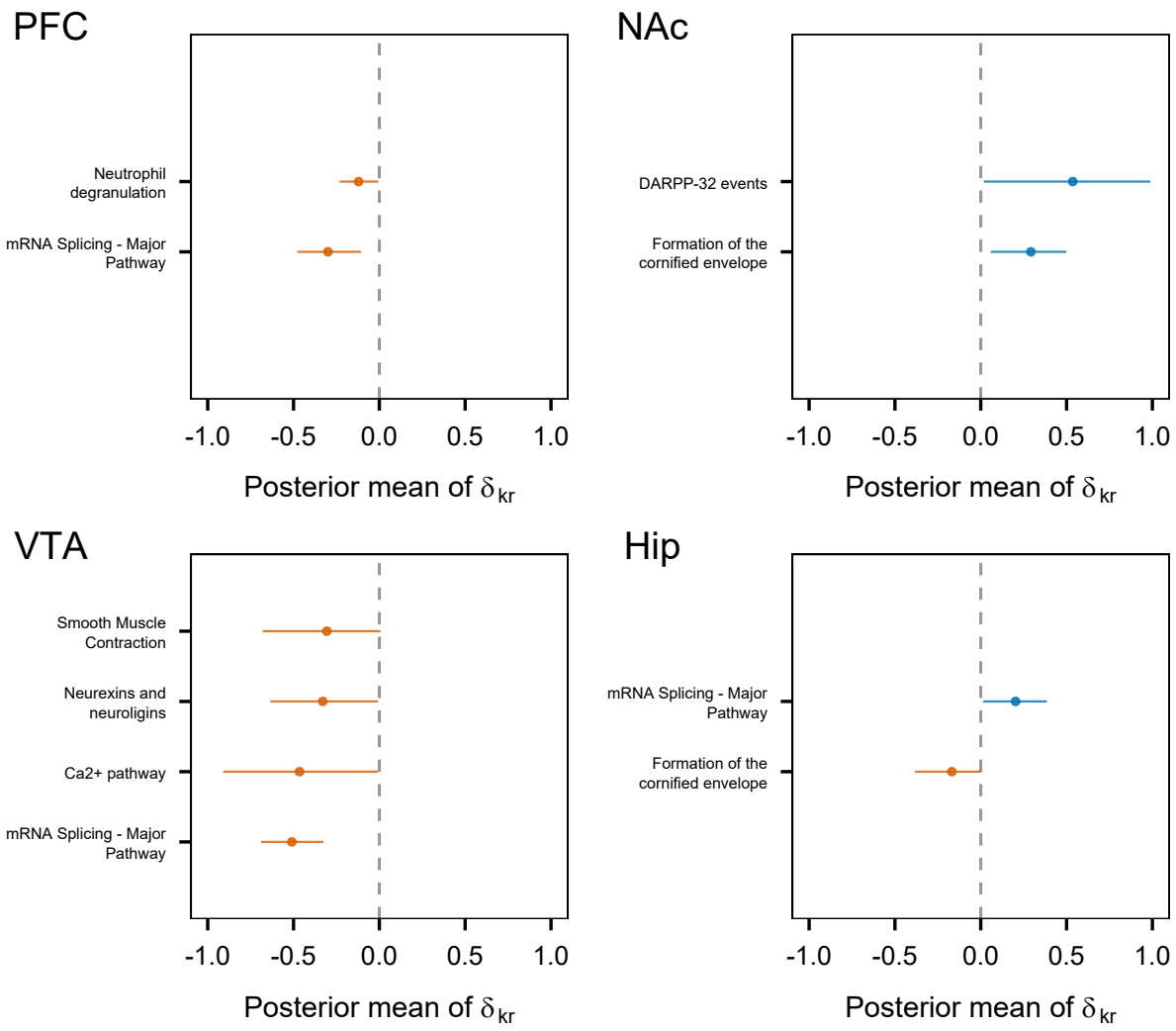


Figure 6: Model 4 estimated pathways per brain region with the largest absolute posterior mean treatment effect and at least 95% posterior probability of being either positive or negative. Horizontal bars show 95% credible intervals. Effect sizes are standardised.

Figure 6 presents the pathways per brain region with the largest absolute treatment effect, as measured by the posterior mean, for Model 4. We only consider pathways which exhibit a 95% probability of being either strictly positive or strictly negative, ensuring that we have substantial posterior confidence in the direction of these large effects. Some pathways are shared across brain regions, such as "mRNA Splicing - Major Pathway" and "Formation of the cornified envelope", although the size and direction of their effects vary. In each brain region, two pathways exhibited large and credible treatment effects, except for VTA, where four negatively affected pathways were found. This agrees with the fact that global shrinkage

was weaker in the VTA compared to the other regions. In general, the posterior means of the largest treatment effects per brain region range between roughly 0.1 and 0.5 in absolute value on the standardised scale. However, when accounting for posterior uncertainty, the 95% credible intervals indicate that treatment could plausibly shift the expected protein intensity by up to one standard deviation for the proteins belonging to that pathway, while holding all other pathway effects constant. For a detailed overview of the treatment effect results using Model 4, see Table 6.

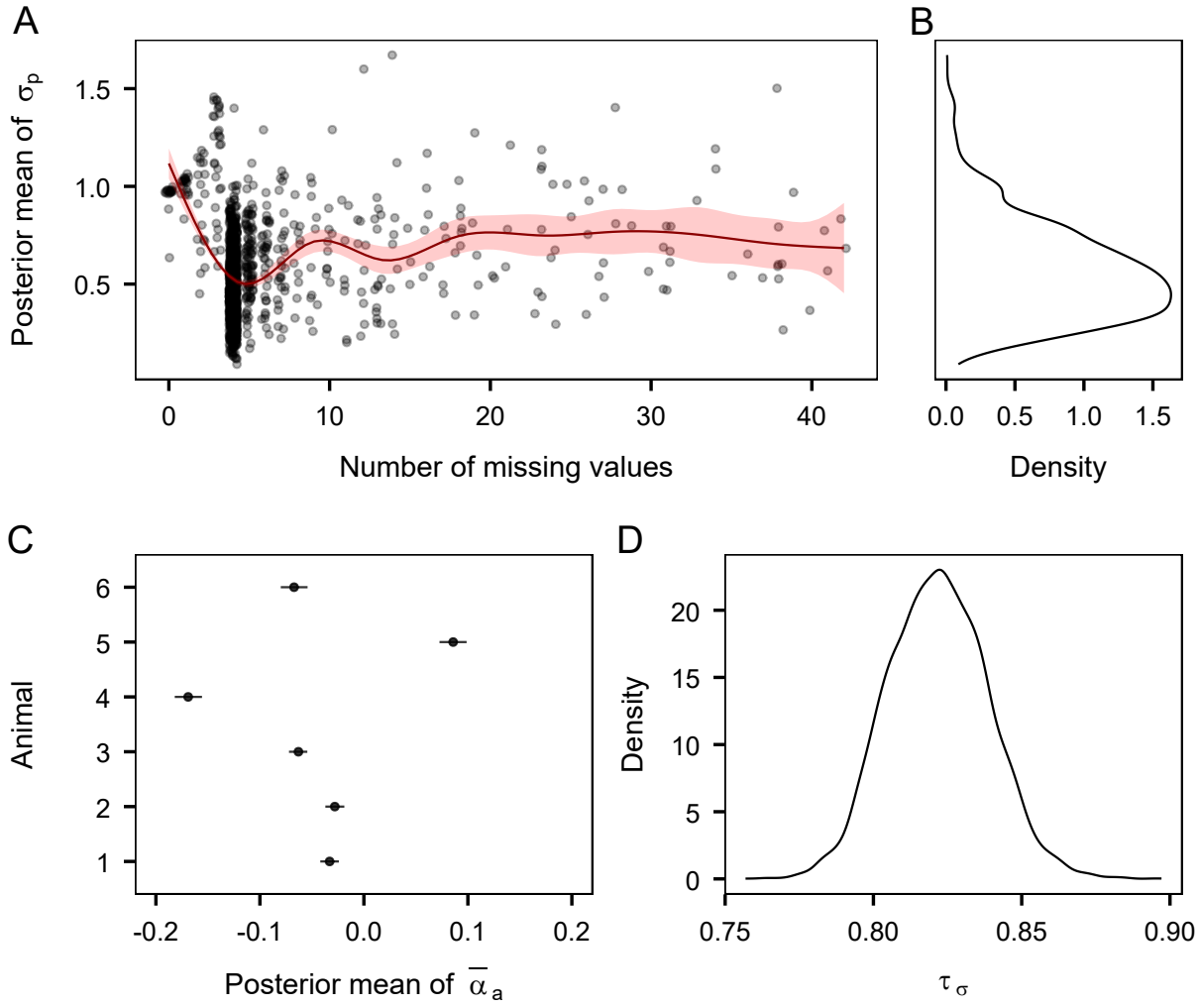


Figure 7: Results for Model 4. **(A):** Number of missing values per protein plotted against the posterior mean values for the per-protein residual standard deviation σ_p. **(B):** Distribution of the posterior mean of σ_p. **(C):** Posterior mean and 95% credible intervals of the baseline intensity per animal aggregated across all proteins, $\bar{\alpha}_a$. **(D):** Posterior distribution of the hierarchical scale parameter τ_σ.

Panel C in Figure 7 shows the posterior mean of the baseline intensity per animal, aggregated

across proteins and brain regions, $\bar{\alpha}_a = \frac{1}{PR}\sum_{p=1}^{P}\sum_{r=1}^{R}\alpha_{apr}$, for Model 4. The plot shows no large differences between animals, indicating that the baseline protein intensity for any specific mouse is effectively indistinguishable from the global average.

The posterior distribution of the protein-specific residual standard deviations σ_p across proteins is shown in Panel B of Figure 7. The Bayesian approximate analogue of the coefficient of determination per protein is $R_p^2 = 1 - \sigma_p^2$, which follows from $R_p^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$ when $SS_{\text{tot}} = 1$, given standardisation of the data (see Section 2). The median implied R_p^2 across proteins is 0.72, indicating that the model on average explains a meaningful share of variation in protein intensity. However, the 2.5th to 97.5th percentile range is $[-0.31, 0.96]$, reflecting heterogeneity in the model's ability to fit different proteins.

Negative implied R_p^2-values, and equivalently values of $\sigma_p > 1.0$ might be explained by the imputation of missing values. While the prior on the missing values is centred around the minimum values for each protein, which lie below zero (see Section 4.1), the likelihood tries to fit the missing values using the expected intensities μ_{ip}. Because the animal-level baseline intensities are small, and the regularised horseshoe prior shrinks most treatment effects toward zero, the expected intensity μ_{ip} will be close to zero for most observations. This discrepancy between the prior and the likelihood for missing values leads to inflated residual standard deviation, which is needed to ensure that the likelihood for the missing values is wide enough to make the imputed values probable.

While the imputation can produce inflated residual variances, the inflation does not increase with higher levels of missingness. As Panel A in Figure 7 shows, the trend for the posterior mean of σ_p is fairly flat, even as the missing values per protein increases to its maximum of 42. This is likely attributed to the partial pooling induced by the hierarchical scale parameter τ_σ. Panel D of Figure 7 shows that τ_σ is well-identified by the data, with its posterior concentrated around $[0.80, 0.85]$, implying moderate regularisation of the per-protein variances toward the prior centre. Without this pooling, inflated σ_p estimates for proteins with many missing values could obscure genuine large treatment effects. A larger σ_p widens and flattens the likelihood, providing weaker evidence that δ_{kr} deviates from zero. This keeps $\tilde{\lambda}_{kr}$ small, leaving δ_{kr} small under the dominant influence of τ_r. This would cause the regularised horseshoe prior to over-shrink, potentially leading to false negatives.

5.1 Comparison to Two-Step Method

The fully Bayesian method and the two-step method largely identify different pathways to be most affected by the treatment. This is illustrated in Figure 8, which shows how the two-step method ranked the seven most credibly affected pathways found by Model 4. The majority of the seven pathways rank below the top 100 pathways per brain region, as measured by FDR shown in parentheses below the rank.

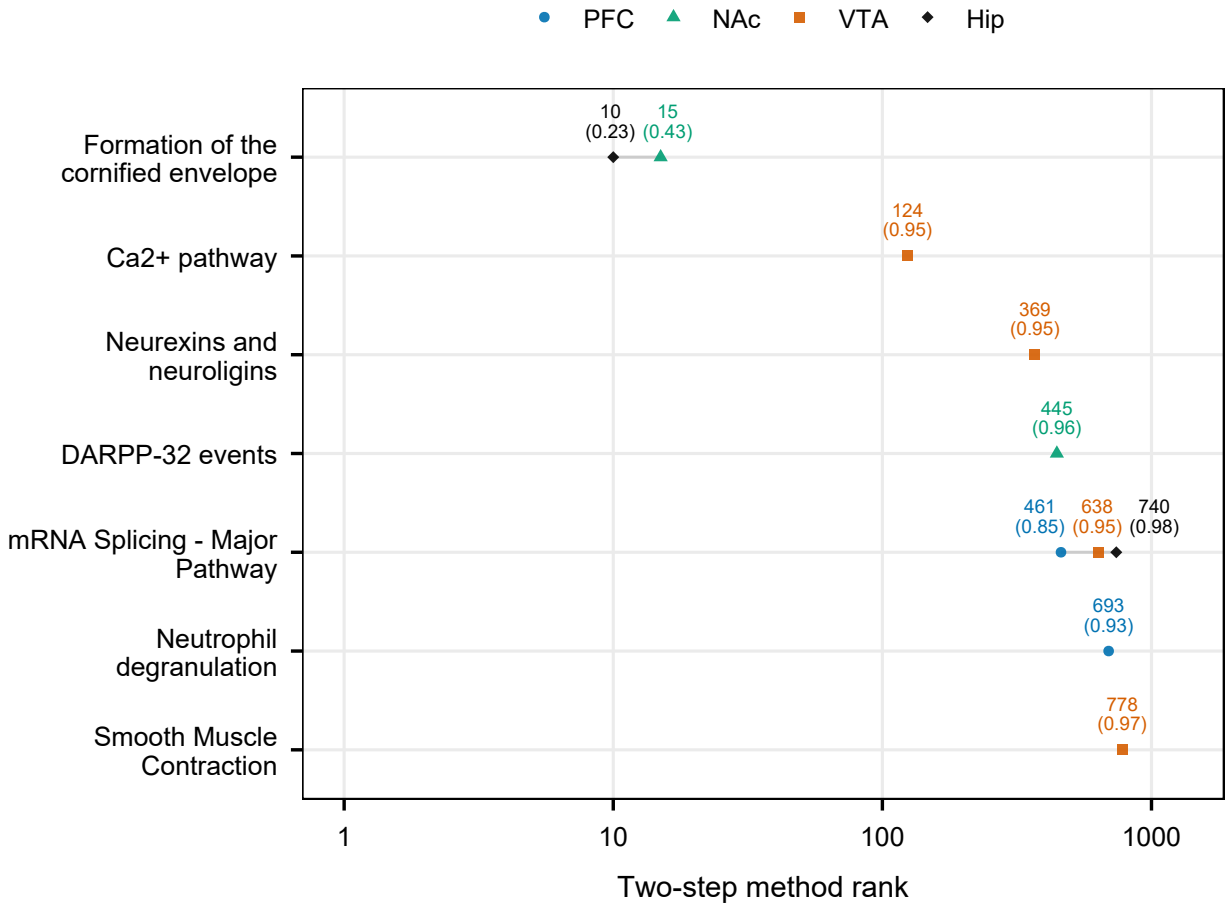


Figure 8: The seven pathways for which Model 4 found the largest credible treatment effects, compared against their ranking by the two-step method (`limma` + `fgsea`). The horizontal axis displays the two-step method's rank on a logarithmic scale. Markers denote the specific brain region, with the False Discovery Rate (FDR) shown in parentheses below the respective rank.

To understand what the two-step method does find, the five most significantly affected pathways per brain region are listed in Table 7 in Appendix C. The pathways are ranked in ascending order by False Discovery Rate (FDR) such that rank 1 denotes the most significant pathway. We see that the two-step method finds very similar top five affected pathways across

PFC, NAc, and VTA, albeit with slightly varying internal order and false discovery rate. The method finds no significantly affected pathways (FDR < 0.05) in the Hippocampus. All the identified pathways involve around 60 proteins, with a median of 58, which is relatively large for a leaf pathway. Across these three brain regions, the two-step method thus repeatedly identifies the same large core biological signal through multiple large pathways, crowding out potential smaller-scale specific signals.

In contrast, Table 6 in Appendix C shows that Model 4 tends to find more granular and region-specific affected pathways, which would appear to be more consistent with the known functional heterogeneity of these brain regions. Model 4 finds treatment effect sizes in the Hippocampus comparable to the other brain regions, suggesting that treatment effects in this region manifest through smaller, more specific pathways that fall below the detection threshold of the two-step method.

One notable similarity between the methods is that "Formation of the cornified envelope" was found by the two-step method at rank 10 and 15 in the Hippocampus and Nucleus Accumbens respectively. However, even though they rank relatively close to the top, with an FDR > 0.20, the pathway is still not found to be significantly affected by the treatment.

These results could be a consequence of the two-step method evaluating each pathway independently and in isolation, without regard to overlap, making inferences susceptible to being dominated by large, overlapping pathways that share the same core biological signal.

6 Conclusion

The primary aim of this study was to determine if pathway analysis could be successfully performed using a fully Bayesian approach, as opposed to the conventional two-step approach which utilises differential expression analysis and enrichment analysis.

By specifying five models of increasing complexity, which can all be fitted to a real-world dataset without convergence issues, we demonstrate that a fully Bayesian approach is a viable alternative for conducting pathway analysis. We show that hierarchical priors, which enable partial pooling of information, improve the out-of-sample predictive performance of the model. In particular, placing hierarchical shrinkage priors on treatment effects has a strong positive impact on this performance, and enables us to single out the pathways that were most strongly affected by the treatment.

Compared to a conventional two-step method, our approach is able to detect more granular and region-specific affected pathways, while the two-step method identifies large similar pathways in three brain regions and finds no affected pathways in the fourth. Furthermore, our approach enables quantification of the treatment effects for each pathway. For the most credibly affected pathways in each brain region, we find that the treatment yields expected posterior mean shifts of up to 0.5 standard deviations, with 95% credible intervals capturing plausible biological effects as large as one standard deviation.

Finally, a challenging factor which limited our ability to model the full pathway biology in the Reactome database was the overlapping and correlated nature of the pathways. Developing methods that can overcome these identifiability issues would add greatly to the utility and applicability of this approach.

Acknowledgements

We thank UPPMAX and the National Academic Infrastructure for Supercomputing in Sweden (NAISS) for providing high-performance computing resources, which greatly helped us fit our models.

References

Alberts, B., Heald, R., Johnson, A., Morgan, D., Raff, M., Roberts, K., Walter, P., Wilson, J. H., & Hunt, T. (2022). *Molecular biology of the cell* (Seventh edition, international student edition). W. W. Norton & Company.

Anderson, N. L., & Anderson, N. G. (1998). Proteome and proteomics: New technologies, new concepts, and new words. *ELECTROPHORESIS*, *19*(11), 1853–1861.

Bauer, S., Gagneur, J., & Robinson, P. N. (2010). GOing Bayesian: Model-based gene set analysis of genome-scale data. *Nucleic Acids Research*, *38*(11), 3523–3532.

Bernardo, J. E. M. (1996). The Concept of Exchangeability and its Applications. *Invited paper for the Bhattacharya Memorial Volume, to be published in the Far East J. Math. Sci.*

Betancourt, M. (2015, June 7). *A Unified Treatment of Predictive Model Comparison*. arXiv: 1506.02273 [stat].

Betancourt, M. (2018). *A conceptual introduction to hamiltonian monte carlo*. arXiv: 1701.02434 [stat.ME]. https://arxiv.org/abs/1701.02434

Betancourt, M., & Girolami, M. (2013, December 3). *Hamiltonian Monte Carlo for Hierarchical Models*. arXiv: 1312.0906 [stat].

Carvalho, C. M., Polson, N. G., & Scott, J. G. (2009). Handling Sparsity via the Horseshoe. *Journal of Machine Learning Research*, *5*.

Casella, G., & Berger, R. L. (2002). *Statistical inference* (2. ed). Duxbury.

Chen, C., Hou, J., Tanner, J. J., & Cheng, J. (2020). Bioinformatics Methods for Mass Spectrometry-Based Proteomics Data Analysis. *International Journal of Molecular Sciences*, *21*(8), 2873.

Efron, B., Tibshirani, R., Storey, J. D., & Tusher, V. (2001). Empirical Bayes Analysis of a Microarray Experiment. *Journal of the American Statistical Association*, *96*(456), 1151–1160.

England, J. L. (2013). Statistical physics of self-replication. *The Journal of Chemical Physics*, *139*(12), 121923.

Gelman, A., Carlin, J. B., Stern, H. S., Dunson, D. B., Vehtari, A., & Rubin, D. B. (2013). Bayesian Data Analysis Third edition. (3).

Hansen, B. E. (2022). *Econometrics*. Princeton University Press.

Hoffman, M. D., & Gelman, A. (2014). The No-U-Turn Sampler: Adaptively Setting Path Lengths in Hamiltonian Monte Carlo. *Journal or Machine Learning Research*, *15*(47), 1593-1623.

Karpievitch, Y., Stanley, J., Taverner, T., Huang, J., Adkins, J. N., Ansong, C., Heffron, F., Metz, T. O., Qian, W.-J., Yoon, H., Smith, R. D., & Dabney, A. R. (2009). A statistical framework for protein quantitation in bottom-up MS-based proteomics. *Bioinformatics*, *25*(16), 2028–2034.

Khatri, P., Sirota, M., & Butte, A. J. (2012). Ten Years of Pathway Analysis: Current Approaches and Outstanding Challenges (C. A. Ouzounis, Ed.). *PLoS Computational Biology*, *8*(2), e1002375.

Klenke, A. (2014). *Probability Theory: A Comprehensive Course*. Springer London.

Kolmogorov, A. N. (1950). *Foundations of the theory of probability*. Chelsea Publishing Company.

Korotkevich, G., Sukhov, V., Budin, N., Shpak, B., Artyomov, M. N., & Sergushichev, A. (2016, June 20). *Fast gene set enrichment analysis*.

Little, R. J. A., & Rubin, D. B. (2020). *Statistical analysis with missing data* (3rd edition). Wiley.

Lönnstedt, I., & Speed, T. (2002). Replicated Microarray Data. *Statistica Sinica*, *12*(1), 31–46.

Milacic, M., Beavers, D., Conley, P., Gong, C., Gillespie, M., Griss, J., Haw, R., Jassal, B., Matthews, L., May, B., Petryszak, R., Ragueneau, E., Rothfels, K., Sevilla, C., Shamovsky, V., Stephan, R., Tiwari, K., Varusai, T., Weiser, J., ... D'Eustachio, P. (2024). The Reactome Pathway Knowledgebase 2024. *Nucleic Acids Research*, *52*(D1), D672–D678.

Neal, R. M. (2011, May 10). MCMC using Hamiltonian dynamics. In *Handbook of Markov Chain Monte Carlo*. CRC Press.

Nelson, D. L., Cox, M. M., Hoskins, A. A., & Lehninger, A. L. (2021). *Lehninger principles of biochemistry* (Eighth edition). Macmillan International Higher Education.

Nezhyva, M., Shahen-Zoabi, S., Kabirova, M., Bentov-Arava, E., Shalev, O., Andrén, P. E., Thornton, C., Yaka, R., Margulis, K., & Jansson, E. T. (2024). Spatial multiomic insights into acute cocaine exposure (E. Klann, Ed.). *PNAS Nexus*, *3*(10), pgae458.

O'Hagan, T. (2004). Dicing with the Unknown. *Significance*, *1*(3), 132–133.

Piironen, J., & Vehtari, A. (2017). Sparsity information and regularization in the horseshoe and other shrinkage priors. *Electronic Journal of Statistics*, *11*(2).

Ritchie, M. E., Phipson, B., Wu, D., Hu, Y., Law, C. W., Shi, W., & Smyth, G. K. (2015). Limma powers differential expression analyses for RNA-sequencing and microarray studies. *Nucleic Acids Research*, *43*(7), e47–e47.

Roberts, G. O., & Rosenthal, J. S. (2004). General state space Markov chains and MCMC algorithms. *Probability Surveys*, *1*.

Schmidt, A., Forne, I., & Imhof, A. (2014). Bioinformatic analysis of proteomics data. *BMC Systems Biology*, *8*(S2), S3.

Schrödinger, E. (1967). *What is Life? The physical aspect of the living cell*. Cambridge University Press.

Schubert, O. T., Röst, H. L., Collins, B. C., Rosenberger, G., & Aebersold, R. (2017). Quantitative proteomics: Challenges and opportunities in basic and applied research. *Nature Protocols*, *12*(7), 1289–1294.

Smyth, G. (2004). Linear Models and Empirical Bayes Methods for Assessing Differential Expression in Microarray Experiments. *Statistical Applications in Genetics and Molecular Biology*, *3*(1), 1–25.

Stan Development Team. (2025). Stan Reference Manual, Version 2.38. https://mc-stan.org/

Strogatz, S. (2024). *Nonlinear Dynamics and Chaos* (3rd ed.).

Subramanian, A., Tamayo, P., Mootha, V. K., Mukherjee, S., Ebert, B. L., Gillette, M. A., Paulovich, A., Pomeroy, S. L., Golub, T. R., Lander, E. S., & Mesirov, J. P. (2005). Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proceedings of the National Academy of Sciences*, *102*(43), 15545–15550.

Tao, T. (2011). *An introduction to measure theory*. American Mathematical Society.

Vehtari, A., Gelman, A., & Gabry, J. (2017). Practical Bayesian model evaluation using leave-one-out cross-validation and WAIC. *Statistics and Computing*, *27*(5), 1413–1432.

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., & Bürkner, P.-C. (2021). Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC. *Bayesian Analysis*, *16*(2).

Voghera, S. (2023). *Bayesian Hierarchical Linear Models for Differential Protein Expression Analysis* [Master's thesis, Uppsala University].

Appendix A Computational Details

Data preparation and post-processing were conducted locally using `R` version 4.5.1. To fit the models, computation was performed on the UPPMAX cluster `Pelle` using `R` version 4.5.1 and `rstan` version 2.32.7. For each model, four Markov chains were run in parallel across four CPU cores.

Table 3: Settings for the NUTS algorithm, and number of parameters.

	Chains	Iterations	`adapt_delta`	`max_treedepth`	Parameters
Model 1	4	2000	0.85	12	41910
Model 2	4	2000	0.85	12	41912
Model 3	4	4000	0.99	12	45160
Model 4	4	4000	0.99	12	45161
Model 5	4	4000	0.99	12	45162

Table 4: Comparison of predictive performance between Model 4 and its extensions. Values represent the difference in estimated ELPD scores relative to the best-performing model.

(a) Comparison to Extension 1

	$\Delta\widehat{\text{elpd}}$	S.E.
Model 4	0.0	0.0
Model 4 (ext. 1)	−98.4	15.6

(b) Comparison to Extension 2

	$\Delta\widehat{\text{elpd}}$	S.E.
Model 4	0.0	0.0
Model 4 (ext. 2)	−42 455.2	291.3

Table 5: Sampling diagnostics. Div = Divergences; Time = sampling time in wall time; ESS = effective sample size.

		$\hat{R}$					Div.	Time	ESS	
		Min	q_{25}	Median	q_{75}	Max	(Count)	(Minutes)	Min	Median
Model 1		0.999	1.000	1.000	1.001	1.017	0	455	449	2852
Model 2		0.999	1.000	1.000	1.001	1.019	0	516	295	3357
Model 3	($p_0 = 3$)	1.000	1.000	1.000	1.000	1.010	0	984	462	6948
Model 4	($p_0 = 3$)	1.000	1.000	1.000	1.001	1.010	0	833	478	7450
Model 4[(1)]	($p_0 = 3$)	1.000	1.000	1.000	1.000	1.010	0	1128	470	7382
Model 4[(2)]	($p_0 = 3$)	1.000	1.000	1.000	1.000	1.010	0	543	629	8818
Model 5	($p_0 = 3$)	1.000	1.000	1.000	1.001	1.013	0	813	473	6894
Model 3	($p_0 = 10$)	1.000	1.000	1.000	1.001	1.014	0	754	396	5845
Model 4	($p_0 = 10$)	1.000	1.000	1.000	1.001	1.019	0	1026	440	6778
Model 5	($p_0 = 10$)	1.000	1.000	1.000	1.001	1.012	0	801	514	6844
Model 3	($p_0 = 20$)	1.000	1.000	1.000	1.001	1.011	0	864	516	7053
Model 4	($p_0 = 20$)	1.000	1.000	1.000	1.001	1.015	0	638	338	5811
Model 5	($p_0 = 20$)	1.000	1.000	1.000	1.001	1.014	0	861	311	6183

[(1)] Extension 1; [(2)] Extension 2.

Appendix B Stan Code

Listing 1: Model 4

```
data {
  int<lower=1> N;  // Number of samples
  int<lower=1> P;  // Number of proteins
  int<lower=1> K;  // Number of pathways

  int<lower=1> N_obs;  // Number of observed values
  vector[N_obs] y_obs;
  array[N_obs] int<lower=1, upper=N> idx_obs_row;
  array[N_obs] int<lower=1, upper=P> idx_obs_col;

  int<lower=1> N_mis;  // Number of missing values
  array[N_mis] int<lower=1, upper=N> idx_mis_row;
  array[N_mis] int<lower=1, upper=P> idx_mis_col;

  vector[P] min_vals;  // Vector of minimum values per protein
  matrix[K,P] X;  // Mapping matrix
  vector[N] w;  // Treatment vector

  int<lower=1> A;  // Number of animals
  array[N] int<lower=1, upper=A> animal_id;

  int<lower=1> R;  // Number of brain regions
  array[N] int<lower=1, upper=R> region_id;

  real<lower=1> p0;  // Prior guess for number of affected pathways
}

transformed data {
  real tau0 = p0 / (K - p0) * 1.0 / sqrt(N);
}

parameters {
  array[A, P, R] real alpha_raw;
  real<lower=0> tau_alpha;
  matrix[K, R] delta_raw;

  vector<lower=0>[R] tau_delta;
  matrix<lower=0>[K, R] lambda_delta;
  real<lower=0> c2;

  vector[P] sigma_raw;
  real<lower=0> tau_sigma;
  vector[N_mis] y_mis;
}

transformed parameters {
  matrix[K, R] lambda_tilde;
```

```
  for (r in 1:R) {
    lambda_tilde[, r] = sqrt(c2) * lambda_delta[, r] ./
                        sqrt(c2 + square(tau_delta[r]) * square(
                           lambda_delta[, r]));
  }

  matrix[K, R] delta;
  for (r in 1:R) {
    delta[, r] = tau_delta[r] * (delta_raw[, r] .* lambda_tilde[, r
       ]);
  }

  matrix[R, P] pathway_shift = delta' * X;

  array[A, P, R] real alpha;
  for (a in 1:A) {
    for (p in 1:P) {
      for (r in 1:R) {
        alpha[a, p, r] = alpha_raw[a, p, r] * tau_alpha;
      }
    }
  }

  vector[N_obs] alpha_obs;
  vector[N_obs] pathway_shift_obs;
  vector[N_mis] alpha_mis;
  vector[N_mis] pathway_shift_mis;
  for (i in 1:N_obs) {
    alpha_obs[i] = alpha[animal_id[idx_obs_row[i]], idx_obs_col[i],
       region_id[idx_obs_row[i]]];
    pathway_shift_obs[i] = pathway_shift[region_id[idx_obs_row[i]],
       idx_obs_col[i]];
  }
  for (i in 1:N_mis) {
    alpha_mis[i] = alpha[animal_id[idx_mis_row[i]], idx_mis_col[i],
       region_id[idx_mis_row[i]]];
    pathway_shift_mis[i] = pathway_shift[region_id[idx_mis_row[i]],
       idx_mis_col[i]];
  }

  vector[P] sigma = exp(sigma_raw * tau_sigma);
}

model {
  // Priors
  to_array_1d(alpha_raw) ~ normal(0, 1);
  to_vector(delta_raw) ~ normal(0, 1);
  sigma_raw ~ normal(0, 1);

  tau_delta ~ student_t(3, 0, tau0);
  to_vector(lambda_delta) ~ student_t(3, 0, 1);
```

```
  c2 ~ inv_gamma(5, 5);

  tau_alpha ~ student_t(3, 0, 1);
  tau_sigma ~ student_t(3, 0, 1);
  y_mis ~ normal(min_vals[idx_mis_col], 1);

  // Likelihood
  y_obs ~ normal(alpha_obs +
                 w[idx_obs_row] .* pathway_shift_obs,
                 sigma[idx_obs_col]);

  y_mis ~ normal(alpha_mis +
                 w[idx_mis_row] .* pathway_shift_mis,
                 sigma[idx_mis_col]);
}

generated quantities {
  vector[N_obs] log_lik;

  for (i in 1:N_obs) {
    log_lik[i] = normal_lpdf(
      y_obs[i] | alpha_obs[i] +
      w[idx_obs_row[i]] * pathway_shift_obs[i],
      sigma[idx_obs_col[i]]
    );
  }
}
```

Appendix C Affected Pathways

Table 6: Model 4. Pathways per brain region with the largest absolute posterior mean treatment effect and $\geq 95\%$ posterior probability of being either strictly positive or strictly negative. Effect sizes are standardised. Size indicates the number of proteins involved in the pathway. CrI = Credible Interval.

Region	**Pathway**	**Size**	**Mean**	**95% CrI**
PFC	mRNA Splicing - Major Pathway	43	-0.299	[-0.478, -0.106]
	Neutrophil degranulation	123	-0.121	[-0.232, -0.007]
NAc	DARPP-32 events	11	0.536	[0.019, 0.988]
	Formation of the cornified envelope	33	0.292	[0.058, 0.499]
VTA	mRNA Splicing - Major Pathway	43	-0.510	[-0.689, -0.326]
	Ca2+ pathway	18	-0.465	[-0.909, -0.009]
	Neurexins and neuroligins	14	-0.330	[-0.636, -0.008]
	Smooth Muscle Contraction	14	-0.306	[-0.680, 0.008]
Hip	mRNA Splicing - Major Pathway	43	0.203	[0.014, 0.385]
	Formation of the cornified envelope	33	-0.168	[-0.383, 0.004]

Table 7: Two-step method (`limma + fgsea`). The five most significant pathways per brain region, ranked by False Discovery Rate (FDR). No significant pathways were found in the Hippocampus (Hip). Size indicates the number of proteins involved in the pathway.

Region	Pathway	Size	FDR	Direction
PFC	Major pathway of rRNA processing in the nucleolus and cytosol	59	1.01×10^{-6}	Down
	Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)	58	1.01×10^{-6}	Down
	PELO:HBS1L and ABCE1 dissociate a ribosome on a non-stop mRNA	57	1.01×10^{-6}	Down
	SRP-dependent cotranslational protein targeting to membrane	58	1.01×10^{-6}	Down
	Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC)	63	1.53×10^{-6}	Down
NAc	Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)	58	3.57×10^{-11}	Down
	ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	58	3.57×10^{-11}	Down
	Nonsense Mediated Decay (NMD) enhanced by the Exon Junction Complex (EJC)	63	5.43×10^{-11}	Down
	PELO:HBS1L and ABCE1 dissociate a ribosome on a non-stop mRNA	57	5.43×10^{-11}	Down
	SRP-dependent cotranslational protein targeting to membrane	58	5.43×10^{-11}	Down
VTA	Major pathway of rRNA processing in the nucleolus and cytosol	59	7.71×10^{-19}	Down
	Nonsense Mediated Decay (NMD) independent of the Exon Junction Complex (EJC)	58	7.71×10^{-19}	Down
	PELO:HBS1L and ABCE1 dissociate a ribosome on a non-stop mRNA	57	7.71×10^{-19}	Down
	ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	58	7.71×10^{-19}	Down
	Formation of a pool of free 40S subunits	63	1.04×10^{-18}	Down
Hip	None < 0.05 FDR	–	–	–

Appendix D Supplementary Material

The data used in this study, along with the full analysis code is available at `https://github.com/FLstats/Master-thesis`.